

RaceDay API Endpoint Plan

Authentication Endpoints

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/auth/register	Register a new user account with role assignment (Participant or Organiser)	None	{ "email": "string", "password": "string", "firstName": "string", "lastName": "string", "phoneNumber": "string", "role": "string" }	201 Created: User registered successfully 400 Bad Request: Validation errors 409 Conflict: Email already exists
POST	/api/auth/login	Authenticate a registered user and return a JWT access token for session management	None	{ "email": "string", "password": "string" }	200 OK: JWT token with user details 401 Unauthorized: Invalid credentials
POST	/api/auth/logout	Invalidate the current user's JWT token to end their session	Any authenticated user	None	200 OK: Logged out successfully 401 Unauthorized: No valid token

User Profile Endpoints

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/users/profile	Retrieve the complete profile information for the currently authenticated user	Any authenticated user	None	200 OK: User profile object 401 Unauthorized: No valid token
PUT	/api/users/profile	Update the profile details for the currently authenticated user	Any authenticated user	{ "firstName": "string", "lastName": "string", "phoneNumber": "string" }	200 OK: Updated user profile 400 Bad Request: Validation errors 401 Unauthorized: No valid token
PUT	/api/users/change-password	Change the password for the currently authenticated user	Any authenticated user	{ "currentPassword": "string", "newPassword": "string" }	200 OK: Password updated successfully 400 Bad Request: Invalid current password

Event Management Endpoints

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events	Retrieve a list of all events with optional filtering by province,	None	Query params: ? province=string&dateFrom=date&dateTo=date&search=string	200 OK: Array of event objects 200 OK:

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
		date range, or search term			Empty array if no events
GET	/api/events/{id}	Return the detailed information for a specific event using the supplied event ID	None	None	200 OK: Complete event object with categories and organiser info 404 Not Found: Event not found
POST	/api/events	Create a new event with its associated categories and registration details	Organiser	{ "name": "string", "description": "string", "location": "string", "province": "string", "startDate": "datetime", "endDate": "datetime", "categories": [{ "categoryId": "int", "entryFee": "decimal", "registrationStart": "datetime", "registrationEnd": "datetime", "maxParticipants": "int" }] }	201 Created: New event object with ID 400 Bad Request: Validation errors 401 Unauthorized: Not logged in 403 Forbidden: Not an organiser
PUT	/api/events/{id}	Update the details of an existing event that belongs to the authenticated organiser	Organiser (owner only)	{ "name": "string", "description": "string", "location": "string", "province": "string", "startDate": "datetime", "endDate": "datetime" }	200 OK: Updated event object 401 Unauthorized: Not logged in 403 Forbidden: Not the event organiser 404 Not Found: Event not found
DELETE	/api/events/{id}	Delete an existing event and all associated data (cascade delete enrolments, results, weather)	Organiser (owner only)	None	200 OK: Event deleted successfully 401 Unauthorized: Not logged in 403 Forbidden: Not the event organiser 404 Not Found: Event not found

Category Management Endpoints

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/categories	Retrieve a list of all available categories in the system	None	None	200 OK: Array of all category objects
POST	/api/categories	Create a new category type that can be used across multiple events	Organiser	{ "name": "string", "description": "string" }	201 Created: New category object 400 Bad Request: Category already exists 401 Unauthorized: Not logged in 403 Forbidden: Not an organiser
GET	/api/events/{eventId}/categories	Retrieve all categories associated with a specific event with their pricing and registration details	None	None	200 OK: Array of category objects with event-specific details 404 Not Found: Event not found
POST	/api/events/{eventId}/categories	Add a new category to an existing event with specific pricing and registration settings	Organiser (owner only)	{ "categoryId": "int", "entryFee": "decimal", "registrationStart": "datetime", "registrationEnd": "datetime", "maxParticipants": "int" }	201 Created: EventCategory object 401 Unauthorized: Not logged in 403 Forbidden: Not event organiser 404 Not Found: Event or category not found
PUT	/api/events/{eventId}/categories/{eventCategoryId}	Update the registration settings (fee, dates, capacity) for a specific category within an event	Organiser (owner only)	{ "entryFee": "decimal", "registrationStart": "datetime", "registrationEnd": "datetime", "maxParticipants": "int" }	200 OK: Updated EventCategory object 401 Unauthorized: Not logged in 403 Forbidden: Not event organiser 404 Not Found: EventCategory not found

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
DELETE	/api/events/{eventId}/categories/{eventCategoryId}	Remove a category from an event (soft delete - maintain historical enrolments)	Organiser (owner only)	None	200 OK: Category removed from event 401 Unauthorized: Not logged in 403 Forbidden: Not event organiser 404 Not Found: EventCategory not found

Event Enrolment Endpoints

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/events/{eventId}/enrol	Enrol a participant in a specific category of an event	Participant	{ "eventCategoryId": "int" }	201 Created: Enrolment object with unique race number 400 Bad Request: Registration closed, category full, or already enrolled 401 Unauthorized: Not logged in 403 Forbidden: Must be a participant
GET	/api/users/enrolments	Retrieve all enrolments for the currently authenticated participant	Participant	None	200 OK: Array of enrolment objects with event and category details 401 Unauthorized: Not logged in 403 Forbidden: Must be a participant
GET	/api/events/{eventId}/enrolments	Retrieve all enrolments for a specific event including participant details	Organiser (owner only)	None	200 OK: Array of enrolment objects with participant details 401 Unauthorized: Not logged in 403 Forbidden: Not event organiser 404 Not Found: Event not found
PUT	/api/enrolments/{enrolmentId}/status	Update the status of a specific enrolment (confirm, owner)	Organiser (event owner)	{ "status": "string", "paymentStatus": "string" }	200 OK: Updated enrolment object 401 Unauthorized: Not logged in

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
		cancel, mark as paid)			403 Forbidden: Not event organiser 404 Not Found: Enrolment not found
DELETE	/api/enrolments/{enrolmentId}	Cancel a participant's enrolment in an event (release capacity)	Participant (self) or Organiser	None	200 OK: Enrolment cancelled successfully 401 Unauthorized: Not logged in 403 Forbidden: Not authorised 404 Not Found: Enrolment not found

Results Management Endpoints

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/events/{eventId}/results	Upload finish times and positions for multiple participants in an event	Organiser (owner only)	{ "results": [{ "enrolmentId": "int", "finishTime": "time", "status": "string" }] }	201 Created: Results uploaded successfully 400 Bad Request: Validation errors 401 Unauthorized: Not logged in 403 Forbidden: Not event organiser
GET	/api/events/{eventId}/results	Retrieve all results for an event with optional category filtering	None	Query params: ? categoryId=int	200 OK: Array of results with participant details, sorted by finish time
GET	/api/users/results	Retrieve all results for the currently authenticated participant	Participant	None	200 OK: Array of results with event and category details 401 Unauthorized: Not logged in 403 Forbidden: Must be a participant
PUT	/api/results/{resultId}	Update a specific result's finish time or position for a participant	Organiser (event owner)	{ "finishTime": "time", "position": "int" }	200 OK: Updated result object 401 Unauthorized: Not logged in 403 Forbidden: Not event organiser 404 Not Found: Result not found

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events/{eventId}/results/leaderboard	Get the event leaderboard with overall and category rankings	None	Query params: ?categoryId=int	200 OK: Leaderboard with positions and finish times

Weather and Route Endpoints

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events/{eventId}/weather	Retrieve weather forecast data for a specific event	None	Query params: ?date=date	200 OK: Array of weather forecast entries 404 Not Found: Event not found
POST	/api/events/{eventId}/weather	Add weather forecast entries for an event	Organiser (owner only)	{ "forecastDateTime": "datetime", "temperature": "float", "condition": "string", "humidity": "int", "windSpeed": "float", "windDirection": "string" }	201 Created: Weather entry object 401 Unauthorized: Not logged in 403 Forbidden: Not event organiser
PUT	/api/events/{eventId}/weather/{weatherId}	Update an existing weather forecast entry	Organiser (owner only)	Same as POST body	200 OK: Updated weather entry 401 Unauthorized: Not logged in 403 Forbidden: Not event organiser
DELETE	/api/events/{eventId}/weather/{weatherId}	Delete a specific weather forecast entry	Organiser (owner only)	None	200 OK: Weather entry deleted 401 Unauthorized: Not logged in 403 Forbidden: Not event organiser
GET	/api/events/{eventId}/route	Retrieve route information for an event (distance, elevation, start/end points)	None	None	200 OK: Route information object
PUT	/api/events/{eventId}/route	Update route information for an event	Organiser (owner only)	{ "distance": "float", "elevation": "float", "startLocation": "string", "endLocation": "string",	200 OK: Updated route information 401

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
				"routeDescription": "string" }	Unauthorized: Not logged in 403 Forbidden: Not event organiser

Summary of Endpoints by Category

Category	Endpoints
Authentication	3 endpoints
User Profile	3 endpoints
Event Management	5 endpoints
Category Management	6 endpoints
Event Enrolment	5 endpoints
Results Management	5 endpoints
Weather and Route	6 endpoints
Total	33 endpoints